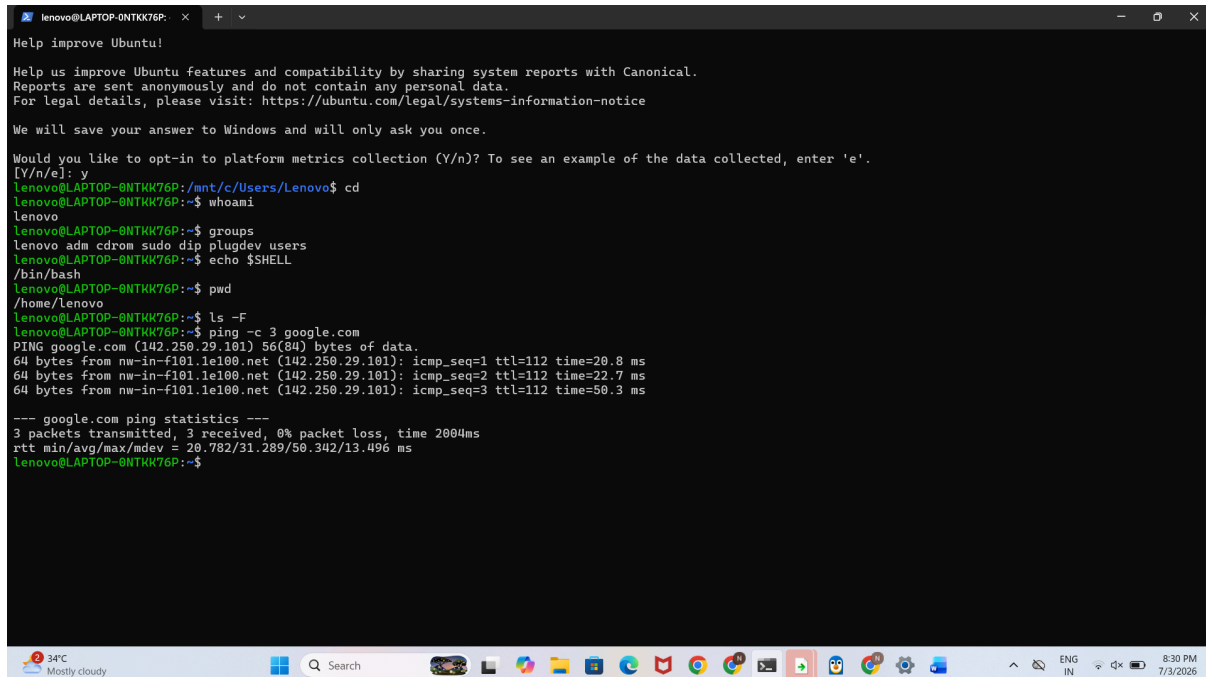


Question 1 Summary: Linux Environment Verification

To verify my Linux environment, I executed a series of diagnostic commands. I used `whoami` to confirm my user identity as `lenovo`, and `groups` to list my security permissions, which confirmed my administrative group membership. I verified my shell environment using `echo $SHELL`, which returned `/bin/bash`. I confirmed my file system location with `pwd` and listed



```
lenovo@LAPTOP-0NTKK76P: ~
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We will save your answer to Windows and will only ask you once.

Would you like to opt-in to platform metrics collection (Y/n)? To see an example of the data collected, enter 'e'.
[Y/n/e]: y
lenovo@LAPTOP-0NTKK76P: /mnt/c/Users/Lenovo$ cd
lenovo@LAPTOP-0NTKK76P:~$ whoami
lenovo
lenovo@LAPTOP-0NTKK76P:~$ groups
lenovo adm cdrom sudo dip plugdev users
lenovo@LAPTOP-0NTKK76P:~$ echo $SHELL
/bin/bash
lenovo@LAPTOP-0NTKK76P:~$ pwd
/home/lenovo
lenovo@LAPTOP-0NTKK76P:~$ ls -F
lenovo@LAPTOP-0NTKK76P:~$ ping -c 3 google.com
PING google.com (142.250.29.101) 56(84) bytes of data:
64 bytes from nw-in-f101.1e100.net (142.250.29.101): icmp_seq=1 ttl=112 time=20.8 ms
64 bytes from nw-in-f101.1e100.net (142.250.29.101): icmp_seq=2 ttl=112 time=22.7 ms
64 bytes from nw-in-f101.1e100.net (142.250.29.101): icmp_seq=3 ttl=112 time=50.3 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 20.782/31.289/50.342/13.496 ms
lenovo@LAPTOP-0NTKK76P:~$
```

files using `ls -F` to ensure a clean workspace. Finally, I utilized `ping -c 3 google.com` to verify that my WSL environment has active network connectivity and that the system is properly configured for internet communication.

Question 2 Summary: Secure Project Workspace Setup

To establish a secure working environment, I created a new directory named `project_workspace`. I inspected its initial permissions using `ls -ld` to establish a baseline. I then applied the `chmod 750` command, which restricts access so that only the owner has full read, write, and execute permissions, while group members have read and execute access, and "others" have zero access. This configuration ensures that sensitive project data is shielded from unauthorized users. Finally, I reviewed the system `umask` to confirm the default permission mask that will govern the security of all future files and directories I create within this session.

```

64 bytes from nw-in-f101.1e100.net (142.250.29.101): icmp_seq=3 ttl=112 time=50.3 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 20.782/31.289/50.342/13.496 ms
lenovo@LAPTOP-0NTKK76P:~$ mkdir project_workspace
lenovo@LAPTOP-0NTKK76P:~$ ls -ld project_workspace
drwxr-xr-x 2 lenovo lenovo 4096 Jul  3 15:04 project_workspace
lenovo@LAPTOP-0NTKK76P:~$ chmod 750 project_workspace
lenovo@LAPTOP-0NTKK76P:~$ ls -ld project_workspace
drwxr-x--- 2 lenovo lenovo 4096 Jul  3 15:04 project_workspace
lenovo@LAPTOP-0NTKK76P:~$ umask
0022
lenovo@LAPTOP-0NTKK76P:~$

```

Question 3 Summary: Hard and Symbolic Links

I created an `original.txt` file and established both a hard link and a symbolic (soft) link. Using the `ls -li` command, I verified that `hard_link.txt` shares the same inode number as the original, meaning it is essentially a second name for the same data. In contrast, `soft_link.txt` has a unique inode and acts as a pointer to the original filename, demonstrating how symbolic links function as shortcuts.

```

0022
lenovo@LAPTOP-0NTKK76P:~$ touch original.txt
lenovo@LAPTOP-0NTKK76P:~$ ln original.txt hard_link.txt
lenovo@LAPTOP-0NTKK76P:~$ ln -s original.txt soft_link.txt
lenovo@LAPTOP-0NTKK76P:~$ ls -li
total 4
44922 -rw-r--r-- 2 lenovo lenovo  0 Jul  3 15:27 hard_link.txt
44922 -rw-r--r-- 2 lenovo lenovo  0 Jul  3 15:27 original.txt
41955 drwxr-x--- 2 lenovo lenovo 4096 Jul  3 15:04 project_workspace
44923 lrwxrwxrwx 1 lenovo lenovo  12 Jul  3 15:28 soft_link.txt -> original.txt
lenovo@LAPTOP-0NTKK76P:~$

```

Question 4 Summary: File Privacy

I created `secret.txt` and checked its initial permissions using `ls -l`. I then applied `chmod 600` to the file, which modifies the permission bits so that only the owner has read and write access, while strictly removing all access for group members and others. This configuration is a critical security practice for ensuring data privacy and protecting sensitive files from unauthorized access.

```
44923 tiwxixixix 1 lenovo lenovo 12 Jul 3 15:28 soft_link.txt -> original.txt
lenovo@LAPTOP-0NTKK76P:~$ touch secret.txt
lenovo@LAPTOP-0NTKK76P:~$ ls -l secret.txt
-rw-r--r-- 1 lenovo lenovo 0 Jul 3 15:33 secret.txt
lenovo@LAPTOP-0NTKK76P:~$ chmod 600 secret.txt
lenovo@LAPTOP-0NTKK76P:~$ ls -l secret.txt
-rw----- 1 lenovo lenovo 0 Jul 3 15:33 secret.txt
lenovo@LAPTOP-0NTKK76P:~$
```

Question 5 Summary: Background Process Management

I initiated a background process using `sleep 100 &`, which allows the command to continue running without blocking the terminal. I verified its active status using `ps aux | grep sleep`, which displays the process ID (PID). After confirmation, I used `kill %1` to terminate the job by its job ID. A final check with `ps aux | grep sleep` confirmed that the process was successfully removed from the system, demonstrating effective background process control.

```
lenovo@LAPTOP-0NTKK76P:~$ sleep 100 &
[1] 1213
lenovo@LAPTOP-0NTKK76P:~$ ps aux | grep sleep
lenovo    1213  0.0  0.0 16112 7788 pts/0    S   15:35   0:00 sleep 100
lenovo    1215  0.0  0.0   4128 2340 pts/0    S+  15:36   0:00 grep --color=auto sleep
lenovo@LAPTOP-0NTKK76P:~$ kill %1
lenovo@LAPTOP-0NTKK76P:~$ ps aux | grep sleep
lenovo    1217  0.0  0.0   4128 2388 pts/0    S+  15:36   0:00 grep --color=auto sleep
[1]+  Terminated                  sleep 100
lenovo@LAPTOP-0NTKK76P:~$
```